MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PRE-BOARD MOCK EXAMINATION – 12 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. The pair of linear equations $2x + 5y = k$ and $kx + 15y = 18$ has infinitely many solutions if: [1]

(A) $k = 3$ (B) $k = 6$ (C) $k = 9$ (D) no value of k exists

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) $k = 6$** [0.5 Mark]

Justification: $\frac{2}{k} = \frac{5}{15} = \frac{1}{3} = \frac{k}{18} \implies \mathbf{k = 6}$ (both equalities hold: $2/6 = 1/3$ and $6/18 = 1/3$). [0.5 Mark]

Q2. If $4x + 6y = 32$ and $4x - 2y = 8$, then the value of $8y$ is: [1]

(A) 24 (B) 16 (C) 32 (D) 8

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 24** [0.5 Mark]

Justification: Subtracting the second equation directly from the first: $(4x - 4x) + (6y - (-2y)) = 32 - 8 \implies 8y = \mathbf{24}$. [0.5 Mark]

Q3. The graphs of $x = 4$ and $y = -3$ intersect at: [1]

(A) $(-3, 4)$ (B) $(4, -3)$ (C) $(0, 0)$ (D) $(4, 0)$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) $(4, -3)$** [0.5 Mark]

Justification: A point on $x = 4$ has x-coordinate 4, and on $y = -3$ has y-coordinate -3; common point is $(\mathbf{4}, \mathbf{-3})$. [0.5 Mark]

Q4. The difference between two numbers is 26 and one number is three times the other. The numbers are: [1]

(A) 39 and 13 (B) 30 and 4 (C) 36 and 10 (D) 42 and 16

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 39 and 13** [0.5 Mark]

Justification: $x - y = 26$ and $x = 3y \implies 3y - y = 26 \implies 2y = 26 \implies y = 13$, $x = 3(13) = \mathbf{39}$. [0.5 Mark]

SECTION B: Assertion-Reason Questions MATHS MANTHAN | (2 × 1 = 2 Marks)

Q5. **Assertion (A):** A two-digit number having tens digit x and units digit y is represented as $10x + y$.
Reason (R): When the digits are reversed, the new number formed is $10y + x$. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B)** [0.5 Mark]

Verification: Both formulas follow from the decimal **place-value rule** (10 × tens + 1 × units). Statement (R) merely states the reversed number and does not explain why the original number is $10x + y$. Hence both (A) and (R) are true, but (R) is not the correct explanation of (A). [0.5 Mark]

Q6. **Assertion (A):** The system of linear equations $x + 2y - 4 = 0$ and $2x + 4y - 12 = 0$ has no solution.
Reason (R): Two parallel lines never intersect each other and hence have no common point of intersection. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: $\frac{1}{2} = \frac{2}{4} \neq \frac{-4}{-12} = \frac{1}{3}$, so lines are parallel and have no solution. Both (A) and (R) are true and (R) explains (A). [0.5 Mark]

SECTION C: Very Short Answer Questions MATHS MANTHAN | (3 × 2 = 6 Marks)

Q7. Determine whether the lines $3x + 2y = 5$ and $2x - 3y = 7$ are consistent or inconsistent. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Ratio Test [2 Marks]

Comparing coefficients: $\frac{a_1}{a_2} = \frac{3}{2}$ and $\frac{b_1}{b_2} = \frac{2}{-3} = -\frac{2}{3}$.

Since $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, the lines intersect at a unique point. **Consistent (unique solution)**

Q8. The sum of two numbers is 85. If their difference is 35, find the numbers. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Forming & Solving Pair [2 Marks]

Let the two numbers be x and y ($x > y$): $x + y = 85$ and $x - y = 35$. Adding: $2x = 120 \implies x = 60$.

Subtracting: $y = 85 - 60 = 25$. **60 and 25**

Q9. The perimeter of an isosceles triangle is 11 cm and its unequal base is 5 cm. Find the equal sides. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Perimeter Formulation [2 Marks]

Let each equal side = x cm and base = y cm: $2x + y = 11$ with $y = 5$.

Substituting $y = 5$: $2x + 5 = 11 \implies 2x = 6 \implies x =$ **3 cm each**.

Q10. Solve for x and y: $mx - ny = m^2 + n^2$ and $x + y = 2m$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Substitution $y = 2m - x$ [1.5 Marks]

From $x + y = 2m$, we get $y = 2m - x$. Substituting into $mx - ny = m^2 + n^2$:

$$mx - n(2m - x) = m^2 + n^2 \implies mx - 2mn + nx = m^2 + n^2 \implies (m+n)x = m^2 + 2mn + n^2 = (m+n)^2$$

Step 2: Solving for x **and** y [1.5 Marks]

Assuming $m + n \neq 0$, dividing by $(m + n)$ gives $x = m + n$. Then $y = 2m - (m + n) = m - n$.

$$\boxed{\mathbf{x = m + n}, \qquad \mathbf{y = m - n}}$$

Q11. The ratio of incomes of two persons is 9 : 7 and the ratio of their expenditures is 4 : 3. If each of them manages to save Rs. 2000 per month, find their monthly incomes. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let incomes be Rs. $9x$, Rs. $7x$ and expenditures be Rs. $4y$, Rs. $3y$.

Formula / Theorem Used: Income − Expenditure = Savings

Step 1: Forming Savings Equations [1.5 Marks]

$9x - 4y = 2000$ — (i) and $7x - 3y = 2000$ — (ii)

Step 2: Eliminating y [1.5 Marks]

Multiplying (i) by 3 and (ii) by 4: $27x - 12y = 6000$ and $28x - 12y = 8000$.

Subtracting first from second: $x = 2000$. Incomes: $9(2000) =$ Rs. 18,000 and $7(2000) =$ Rs. 14,000.

Monthly Incomes: Rs. **18,000** and Rs. **14,000**

Q12. A two-digit number is 4 times the sum of its digits and twice the product of its digits. Find the number. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let tens digit $= x$ $(x \neq 0)$ and units digit $= y$; number $= 10x + y$.

Step 1: Using Sum-of-Digits Relation [1.5 Marks]

$10x + y = 4(x + y) \implies 10x + y = 4x + 4y \implies 6x = 3y \implies y = 2x$ — (i)

Step 2: Using Product-of-Digits Relation [1.5 Marks]

$10x + y = 2xy$. Substituting $y = 2x$: $10x + 2x = 2x(2x) \implies 12x = 4x^2 \implies 4x(x - 3) = 0$.

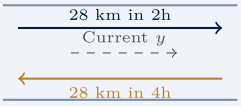
Since $x \neq 0$, we get $x = 3$, and $y = 2(3) = 6$. Original number $= 10(3) + 6 =$ **36**.

Q13. A sailor can row a boat at speed x km/h in still water and the river flows at speed y km/h. The boat travels 28 km downstream in 2 hours and takes 4 hours to return the same distance upstream. Formulate a pair of linear equations in x and y, find the speed of the sailor in still water and the speed of the current, and find the total time taken to row 35 km in still water. [5]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Formula / Theorem

Used: Downstream $= x + y$, Upstream $= x - y$, $t = \dfrac{D}{v}$



Step 1: Forming Equations [2 Marks]

Downstream: $2(x + y) = 28 \implies x + y = 14$ — (i)

Upstream: $4(x - y) = 28 \implies x - y = 7$ — (ii)

Step 2: Solving for x and y [2 Marks]

Adding (i) and (ii): $2x = 21 \implies x = \frac{21}{2} = 10.5$ km/h. From (i): $y = 14 - 10.5 = 3.5$ km/h.

Step 3: Time to Row 35 km in Still Water [1 Mark]

$t = \dfrac{35}{10.5} = \dfrac{350}{105} = \dfrac{10}{3}$ hours $= 3$ hours 20 minutes.

$$\boxed{x = \mathbf{10.5}\text{ km/h}, \quad y = \mathbf{3.5}\text{ km/h}, \quad t = \mathbf{3}\text{ h }\mathbf{20}\text{ min } \left(\frac{\mathbf{10}}{\mathbf{3}}\text{ h}\right)}$$

Q14. CASE STUDY: Railway Reservation Tariff & Concession Rules

Under Indian Railways passenger tariff rules, a child aged between 5 and 11 years is eligible for a half-fare ticket. However, the full reservation surcharge and superfast charge apply equally to both adult (full) and child (half) tickets. A passenger booked 1 full adult first-class AC ticket from New Delhi to Varanasi and paid Rs. 2530 ($F + R = 2530$). On another booking for the same journey, a family booked 1 full ticket and 1 half-fare ticket and paid a total sum of Rs. 3810 ($1.5F + 2R = 3810$). [4]

(i) Formulate the pair of linear equations in basic fare F and reservation charge R. [1]

(ii) Determine the full basic first-class AC fare (F). [1]

(iii) Determine the fixed reservation charge per ticket (R). [1]

(iv) Calculate the total ticket price for a family consisting of 2 adults and 2 children travelling on the same route. [1]

— OR —

Find the total amount saved by booking 1 half ticket instead of 1 full ticket. [1]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Sub-part (i): Tariff Equations [1 Mark]

$F + R = 2530$ — (i) and $1.5F + 2R = 3810$ (or $3F + 4R = 7620$) — (ii)

Sub-part (ii): Full Basic Fare (F) [1 Mark]

Multiplying (i) by 3: $3F + 3R = 7590$. Subtracting from $3F + 4R = 7620$: $R = 30$. Thus $F = 2530 - 30 =$ Rs. **2,500**.

Sub-part (iii): Fixed Reservation Charge (R) [1 Mark]

From the elimination step above: **R = Rs. 30**.

Sub-part (iv): Total Price for 2 Adults and 2 Children [1 Mark]

Adult ticket $= F + R = 2530$; Child ticket $= \frac{1}{2}F + R = 1250 + 30 = 1280$.

Total $= 2(2530) + 2(1280) = 5060 + 2560 =$ Rs. **7,620**.

Sub-part (iv) [OR Option]: Amount Saved on 1 Half Ticket vs. 1 Full Ticket [1 Mark]

Savings = Full Ticket − Half Ticket $= 2530 - 1280 =$ Rs. **1,250**.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★